

Problem1.**1.Methods analysis**

CLIP 主要透過兩個模型來運作，分別是 image encoder 和 text encoder，而在 CLIP 的訓練上主要以一張圖片對應一個句子的方式做訓練，並採用了大量的資料集，以 OpenAI-CLIP 為例則用了 4 億筆訓練資料，相較於傳統的圖片對應 label 的資料集，一個句子更可以學到該種 label 的不同特徵，另外大量的預訓練資料亦可包含大量不同 label 的圖片。

2. Prompt-text analysis

My model:ACC=0.71

i. This is a photo of {object}:ACC=0.6092

ii. This is a {object} image.:ACC=0.6836

iii. No {object}, no score.:ACC=0.564

I also try different prompt-text below.

this is a photo of a {object}:ACC=0.6868

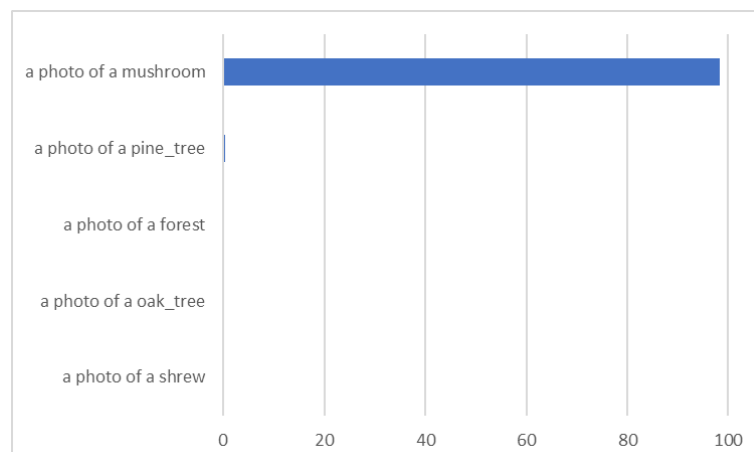
a {object} image:ACC=0.6836

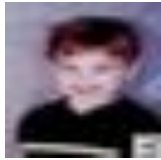
{object}:ACC=0.6352

因為在預訓練 CLIP 時，所輸入的文字為 a photo of {object}，所以在 predict 時如輸入的 prompt-text 和 a photo of {object}越像則 performance 越好，至於為何 this is a photo of a {object}的 performance 會大於 This is a photo of {object}，我認為則是預訓練時的資料多數為 a photo of a {object}，再加上圖片多為單一物體，所以 this is a photo of a {object}的 performance 會更好，另外由於預訓練時文字為一個名詞，因此若預測時 prompt-text 為名詞時則會比 prompt-text 為一句子時表現普遍更佳。

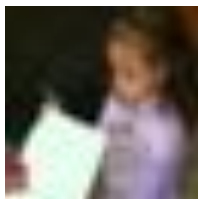
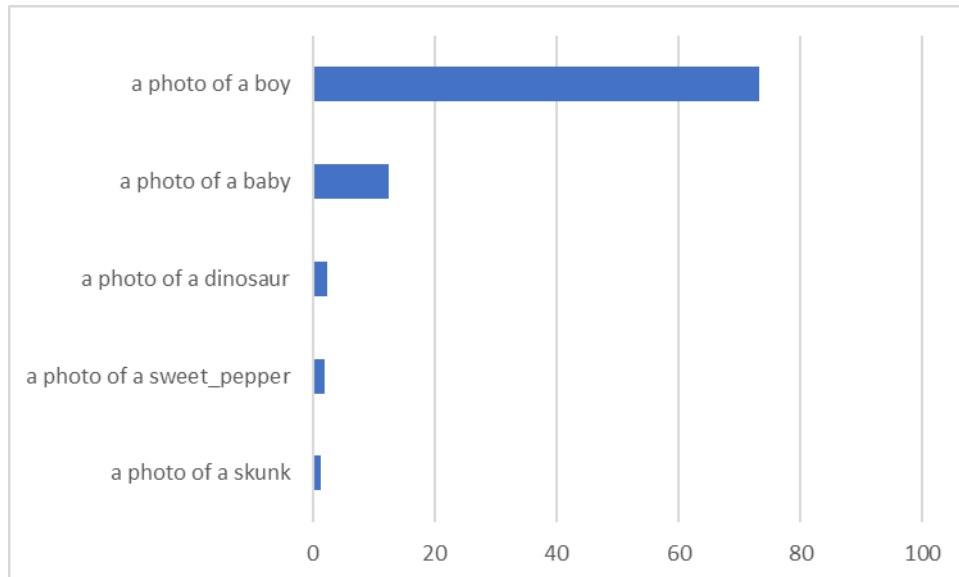
3.Quantitative analysis

correct label: mushroom, correct probability: 98.29%

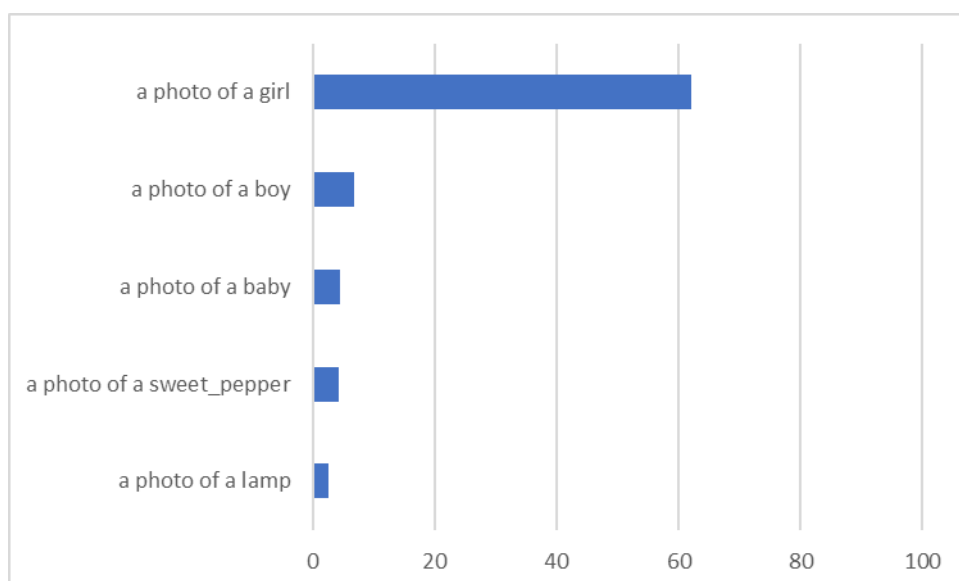




correct label: boy, correct probability: 73.19%



correct label: girl, correct probability: 62.11%



Problem2.

1.

setting:

criterion: torch.nn.CrossEntropyLoss()

optimizer: torch.optim.Adam(model.parameters(), lr=2e-5, betas=(0.9, 0.98), eps=1e-9)

epochs: 30

encoder: vit_base_patch_16_384

decoder_layer:

torch.nn.TransformerDecoderLayer(d_model=384,nhead=12,dim_feedforward=2048)

decoder: torch.nn.TransformerDecoder(decoder_layer, num_layers=4)

CIDEr:0.76521 CLIPScore:0.69924

2.I try different models, mainly on model architecture modifying. Other settings is same to the best model.

i.

encoder: vit_base_patch_16_224_in21k

decoder_layer:

torch.nn.TransformerDecoderLayer(d_model=256,nhead=8,dim_feedforward=2048)

decoder: torch.nn.TransformerDecoder(decoder_layer, num_layers=6)

CIDEr:0.43153 CLIPScore:0.65641

ii.

encoder: vit_base_patch_16_384

decoder_layer:

torch.nn.TransformerDecoderLayer(d_model=256,nhead=8,dim_feedforward=2048)

decoder: torch.nn.TransformerDecoder(decoder_layer, num_layers=4)

CIDEr:0.48396 CLIPScore:0.66844

iii.

encoder: vit_large_patch_16_384

decoder_layer:

torch.nn.TransformerDecoderLayer(d_model=384,nhead=12,dim_feedforward=2048)

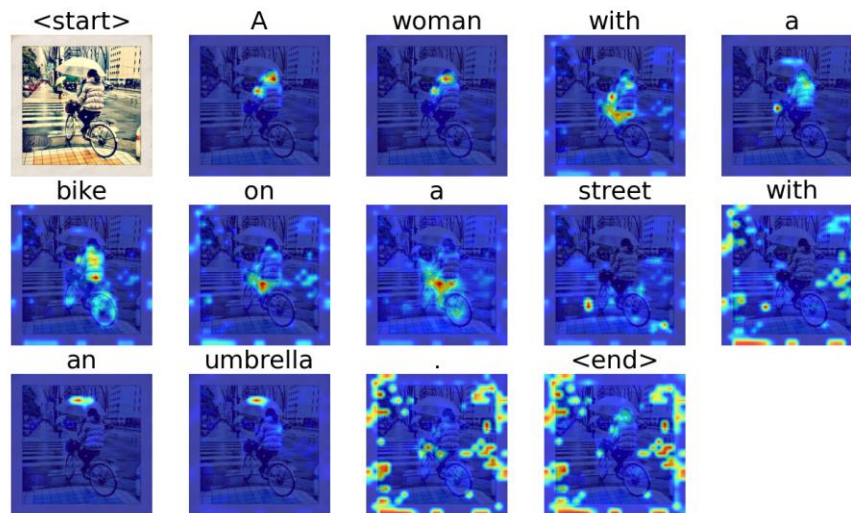
decoder: torch.nn.TransformerDecoder(decoder_layer, num_layers=4)

CIDEr:0.78382 CLIPScore:0.69264

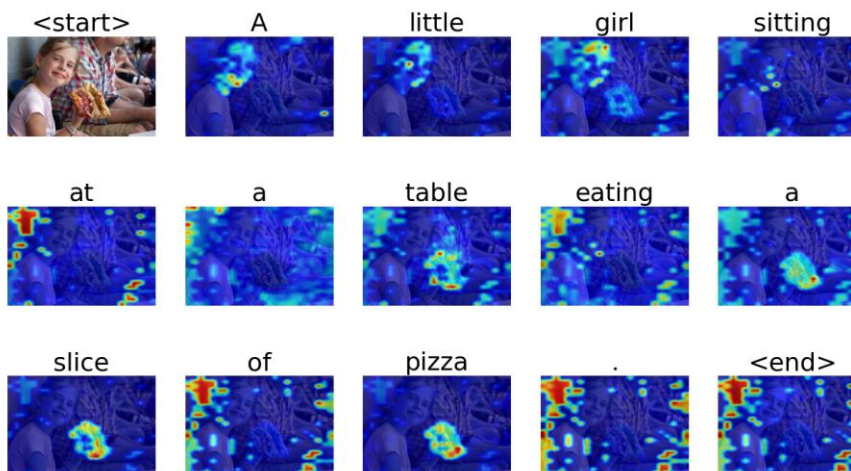
Problem3:

1.

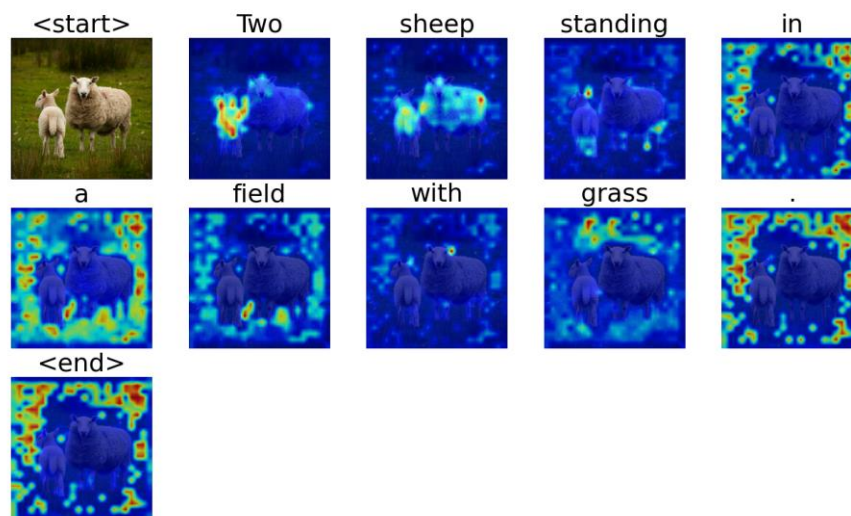
i.bike.png



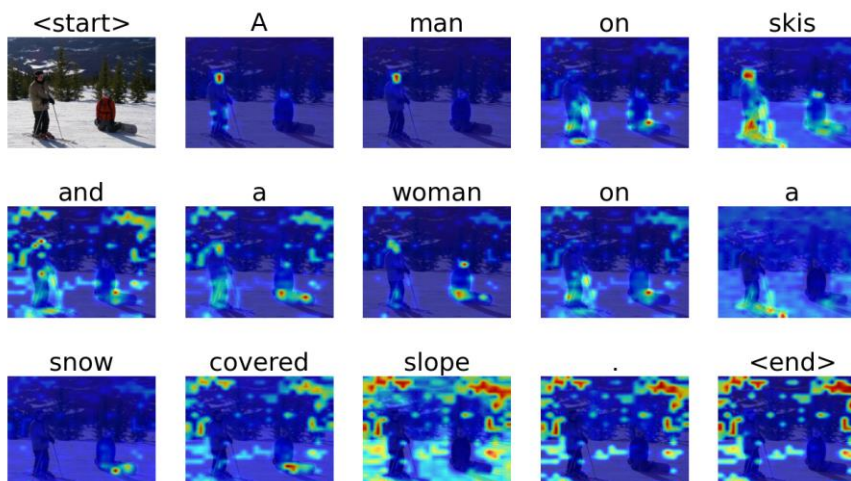
ii.girl.png



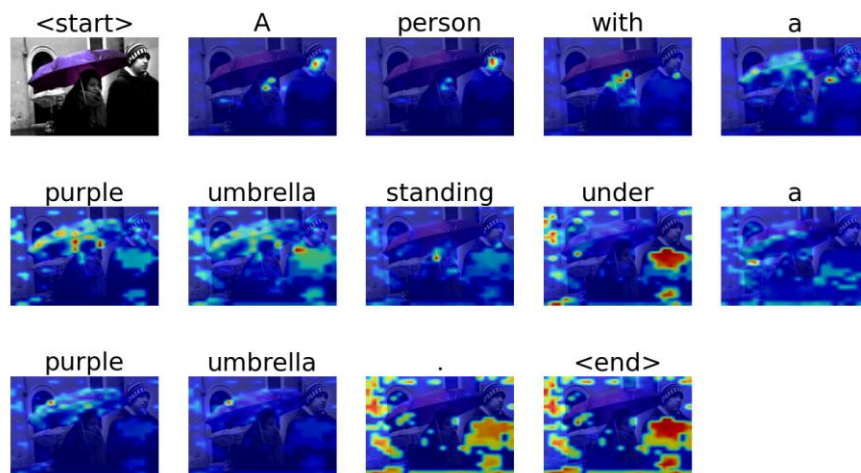
iii.sheep.png



iv.ski.png

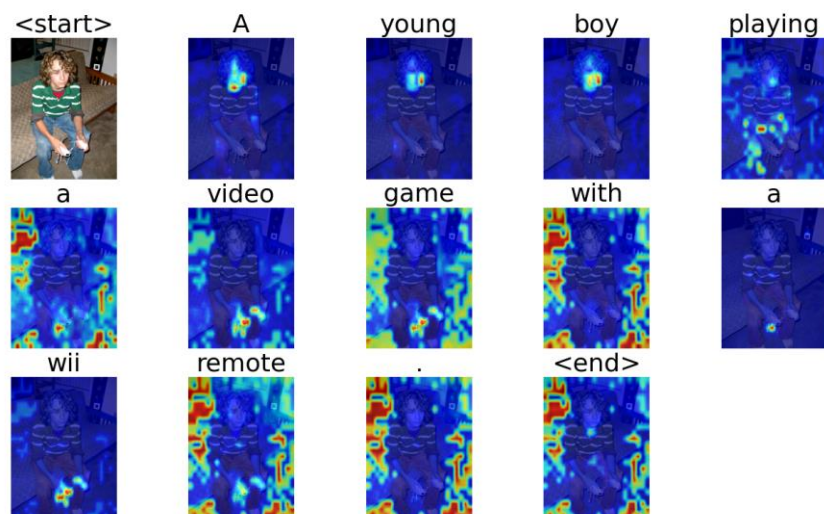


v.umbrella.png

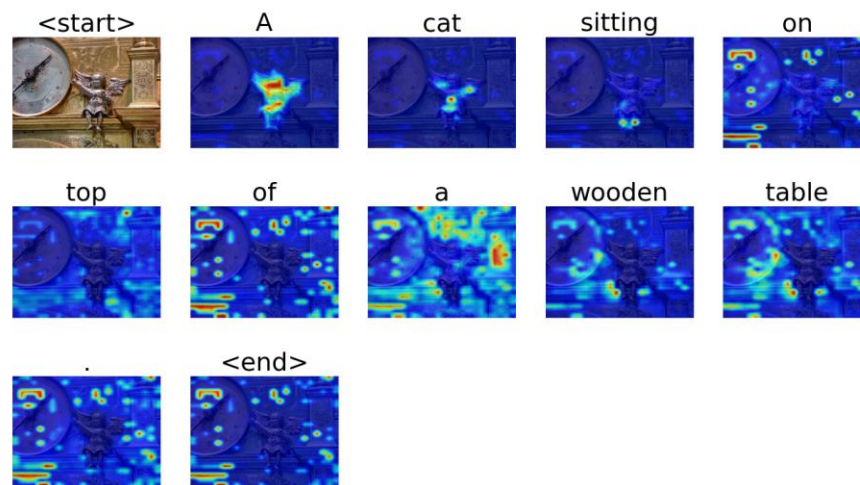


2.

top-1



last-1



3.Yes the captions is roughly reasonable.對於名詞或動詞則多數位於正確位置上，可以改善的地方是在判斷該物體或是動作，至於形容詞、介係詞或是量詞則 attention map 較難顯示出來，而句號和<end>對應的 attention map 相同則為預期中的結果。